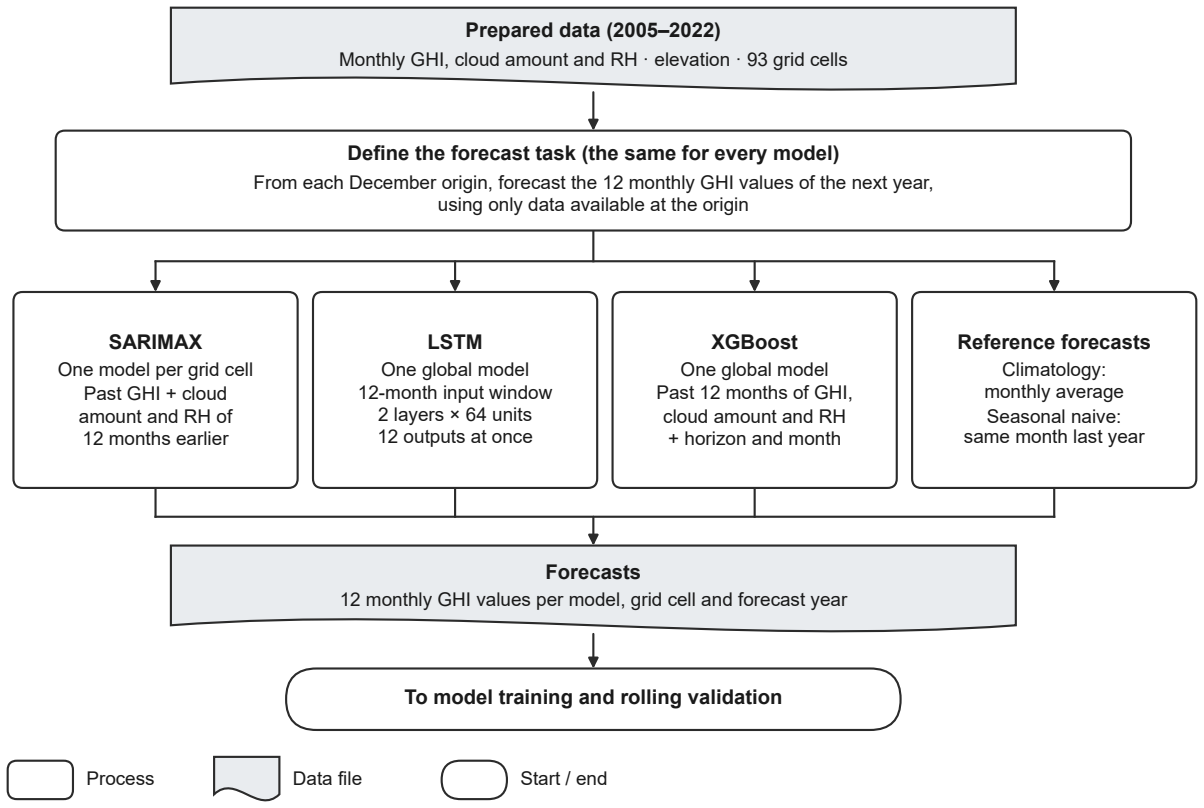

Prepared data (2005–2022)
Monthly GHI, cloud amount and RH · elevation · 93 grid cells
Define the forecast task (the same for every model)
From each December origin, forecast the 12 monthly GHI values of the next year, using only data available at the origin
SARIMAX
One model per grid cell
Past GHI + cloud amount and RH of 12 months earlier
LSTM
One global model
12-month input window
2 layers × 64 units
12 outputs at once
XGBoost
One global model
Past 12 months of GHI, cloud amount and RH + horizon and month
Reference forecasts
Climatology: monthly average
Seasonal naive: same month last year
Forecasts
12 monthly GHI values per model, grid cell and forecast year
To model training and rolling validation
Process
Data file
Start / end